Adaptive Agentic RAG Financial Research Assistant.

1) Client (POST /finance-agent)

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2) FASTAPI Web Layer

(Request Validation,
Session memory management)

* Fast text memory context

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3) Agentic RAG Orchestrator (Document loading → Chunking → Embedding → Vector DB (Chroma DB))

Meta (Additional Information)

Understanding Query → Query Classification

Multi-Hop Reasoning, Comparison, Fact Retrieval, Decision Support

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Query Rewriting

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Multi-Hop Query Decomposition
(Decompose the query into logical sub-queries)

Chunk

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4) Hybrid Retrieval

- Dense - Chroma DB
- Sparse - BM25
- Fusion - Reciprocal Rank Fusion (RRF)

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5) Sufficiency and Deep Search Mode

Triggers upto 2 if low confidence

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6) Persona Based Response, Faithfulness verification, Confidence scoring (0-1) Eg: 0.8

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7) JSON API Response

(Answer) Result, Output..., etc...